

MODULE 15

Database Normalization 1NF 2NF 3NF

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

Database Normalization 1NF 2NF 3NF

What is Normalization?

Organising data to reduce redundancy and improve integrity.

Splits large tables into smaller related ones.

Normal forms: 1NF, 2NF, 3NF, BCNF.

First Normal Form 1NF

Each column has atomic (indivisible) values.

No repeating groups.

Wrong:

id | name | courses

1 | Arshith | SQL, Python

Right 1NF:

id | name | course

1 | Arshith | SQL

1 | Arshith | Python

Second Normal Form 2NF

1NF + no partial dependencies.

Wrong: order_id, product_id | qty | product_name

(product_name depends only on product_id)

Fix: split into Orders(order_id, product_id, qty)

and Products(product_id, product_name)

Third Normal Form 3NF

2NF + no transitive dependencies.

Wrong: student_id | dept_id | dept_name | dept_head

(dept_name depends on dept_id, not student_id)

Fix: Students(student_id, dept_id)

and Departments(dept_id, dept_name, dept_head)

Denormalization

Adding redundancy intentionally for read performance.

When: read-heavy apps, analytics, costly JOINS.

Balance: normalize for data integrity,
denormalize for read speed.